

Resilience Analysis in Off-Grid LoRa Mesh Networks: Evaluation of Meshtastic Profiles in Long-Range Propagation Scenarios

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Abstract. The integration of LoRa technologies with mesh topologies represents a robust alternative for off-grid communications in emergency scenarios within smart cities. Meshtastic firmware implements a decentralised mesh network over LoRa where each node acts simultaneously as end device and router, enabling communication via Bluetooth-connected mobile devices without reliance on conventional infrastructure. Within the Colombian context (915 MHz ISM band), this work establishes design and planning criteria through a controlled guided-link methodology that isolates the LoRa physical layer from propagation effects, enabling deterministic characterisation of all eight Meshtastic modem presets at three transmission power levels (42 datasets). The results reveal a performance partitioning governed primarily by Spreading Factor (SF): *Short* presets (SF7–SF8) fail at 110 dB to 120 dB of path attenuation, *Medium* presets (SF9–SF10) sustain links up to 135 dB to 150 dB, and *Long* presets (SF11–SF12) maximise coverage, with *Long Slow* reaching 180 dB before failure—a 60 dB to 70 dB advantage over the fastest profiles. The SNR analysis confirms sub-noise-floor demodulation down to -18 dB for SF12, with abrupt link failure occurring within 2 dB to 4 dB of the theoretical limit. Based on these thresholds, three operational regimes are defined—high-density IoT, balanced urban mesh, and maximum-range emergency—providing network designers with quantitative criteria to select the appropriate configuration and node density for smart city deployments.

Keywords: Meshtastic · LoRa · Mesh networks · Emergency communications · Smart city · Off-grid · Routing · SX1262

1 Introduction

The transformation of urban environments into smart cities fundamentally depends on the ability to deploy communication networks that connect thousands of devices with limited energy, memory, and processing resources. In this ecosystem, Low Power Wide Area Networks (LPWAN) based on LoRa (Long Range)

modulation have emerged as a technically compelling solution, filling the gap between short-range networks such as WiFi and traditional cellular infrastructure. Through its Chirp Spread Spectrum (CSS) modulation, LoRa provides superior resistance to interference and multipath fading, which is critical for data transmission in dense metropolitan environments [1].

Traditionally, these solutions have been deployed under the LoRaWAN protocol using a star topology, where end devices communicate with centralised gateways connected to Internet backhaul. However, disaster events—earthquakes, floods, massive power outages, or cyberattacks on critical infrastructure—can disable the 4G/5G cellular and fibre optic networks that support urban services such as transportation, energy distribution, and public safety [8]. This reality has driven the development of decentralised mesh network architectures capable of maintaining communication continuity without any fixed infrastructure, a factor identified as a fundamental pillar of urban resilience in the context of disaster-resilient smart city frameworks [5, 12].

The Meshtastic firmware implements precisely this approach over LoRa, enabling every node to function simultaneously as an end device and a router through a managed flooding protocol combined with CSMA/CA. By enabling Bluetooth Low Energy (BLE) connectivity, users can communicate through Meshtastic nodes using their mobile devices without depending on conventional cellular or Internet infrastructure. Beyond emergency messaging, the protocol natively supports telemetry data transmission from IoT sensors, while compatibility with MQTT allows Internet-connected nodes to act as gateways to urban management platforms such as ThingsBoard or Grafana [13]. This hybrid architecture enables autonomous operation during infrastructure failures, bridging the gap between emergency communications and smart city services such as waste management, smart metering, and real-time structural health monitoring of bridges and critical infrastructure.

In the context of Latin American smart cities, where telecommunications infrastructure can be heterogeneous and vulnerable, the ability to operate in the 915 MHz ISM band without requiring a licence, combined with low-cost hardware and solar-powered autonomous operation, makes Meshtastic a particularly compelling solution for community early warning systems. The present work seeks to establish a design and action plan for the use of Meshtastic mesh networks in the 915 MHz frequency band, with the goal of contributing to the development of resilient, low-cost communication infrastructure for smart urban environments.

The objective of this work is to establish design and planning criteria for Meshtastic networks operating in the 915 MHz ISM band. To this end, this paper makes the following contributions:

1. A controlled guided-link experimental methodology that isolates LoRa physical layer behaviour from propagation effects, enabling deterministic sensitivity and PER characterisation.
2. A comprehensive evaluation of all eight Meshtastic modem presets (*Short Turbo* through *Long Slow*) at three transmission power levels, yielding 42 datasets that map the operational envelope of the SX1262 transceiver.

3. Spectral verification of LoRa emissions using both SDR-based and calibrated spectrum analyser measurements, confirming CSS modulation compliance in the 915 MHz band.
4. Practical design guidelines for coverage planning in urban emergency and IoT deployment scenarios, derived from empirical sensitivity thresholds.

The remainder of this paper is organised as follows. Section 3 details the experimental setup, hardware characterisation, and test matrix. Section 4 presents the measurement results and sensitivity analysis. Section 5 discusses the implications for configuration selection and node density planning. Finally, Section 6 summarises the findings and outlines future work.

2 State of the art

The development of smart cities requires communication technologies that achieve a balance between long range and low energy consumption, especially when connecting devices with limited resources. In this context, LoRa (Long Range) has established itself as one of the most relevant solutions at the physical layer, thanks to its Chirp Spread Spectrum (CSS) modulation, which allows it to operate robustly against interference and fading effects typical of dense urban environments. This characteristic translates into high receiver sensitivity, making it possible to establish links where other wireless technologies are not viable. A key aspect of these networks is energy efficiency. Mechanisms such as Channel Activity Detection (CAD) allow devices to identify channel activity without keeping the receiver continuously active, thereby reducing battery consumption. However, parameters such as the Spreading Factor (SF) and bandwidth (BW) not only determine communication range, but also the time on air (airtime), which directly impacts node battery life.

In relation to this, studies such as that of [1] analyze in detail the behavior of the LoRa physical layer, establishing the mathematical foundations for understanding how SF and BW influence metrics such as bit rate and symbol duration. Their results show that SF is the most determining factor in coverage, which is consistent with experimental observations in practical configurations such as those used in Meshtastic.

On the other hand, [4] evaluate LoRa performance in environments with adverse propagation conditions, such as university campuses with multiple obstacles. Their results show how SF and Code Rate (CR) affect packet loss rate, providing a relevant experimental basis for analysis in dense urban scenarios. Along the same lines, [13] present real-world implementations of private LoRa networks, reporting ranges of up to 7.5 km in urban environments using high SF configurations, which serves as a reference for long-range propagation studies.

At the architecture level, most current deployments are based on LoRaWAN, which uses a star topology dependent on gateways and central servers. Nevertheless, alternatives such as mesh networks have emerged, enabling direct communication between nodes without the need for fixed infrastructure. This approach, adopted by solutions such as Meshtastic, is particularly attractive in scenarios

where traditional connectivity is unavailable or may fail, such as in emergency situations.

The behavior of these networks under real conditions has been widely studied. [2] analyzed the operation of The Things Network over several months, identifying issues associated with packet collisions and SF usage distribution in congested networks. Likewise, [11] demonstrate the applicability of LoRa in critical infrastructure monitoring, evaluating quality of service parameters such as latency and throughput, which reinforces its usefulness in urban environments.

From a broader perspective, systematic reviews such as that of [8] show the growing adoption of LoRa in sectors such as agriculture, mining, and early warning systems. At the same time, [6] highlight important challenges related to security and interconnectivity in LoRaWAN architectures, which opens the door to decentralized solutions such as mesh networks. Regarding the regulatory framework, [5] emphasize the importance of international standards such as ISO 37120 and IEEE P2784, which promote interoperability and scalability in smart city development.

Finally, the deployment of these technologies is often supported by controlled testing environments. [7] introduce the concept of Living Labs, which allows solutions to be validated before large-scale implementation. In a complementary manner, studies such as those of [12] and [3] emphasize the importance of integrating sensors and resilient networks to improve key aspects such as urban mobility and public safety.

Taken together, the reviewed literature shows that the combination of LoRa with mesh architectures represents a solid alternative for the development of resilient networks, especially in scenarios where centralized infrastructure is not viable. This approach is particularly relevant in emergency communication applications and off-grid environments, where service continuity is a critical requirement.

3 Methodology

The design of wireless networks using Meshtastic nodes requires a rigorous understanding of radio frequency (RF) parameters and hardware capabilities. This section details the experimental framework, the characterisation of the devices under test (DUT), and the physical layer principles governing system performance.

3.1 Experimental Setup and System Architecture

To evaluate link resilience and packet error rate (PER) deterministically, the RF link was replaced by a guided medium. This approach eliminates stochastic variables such as multipath fading, environmental interference, and temporal variability of the wireless channel, providing a controlled and repeatable environment for sensitivity analysis.

Figure 1 illustrates the experimental link model. The antennas were substituted by a wired connection where the RF channel is emulated through a cascade of fixed attenuators (At_{1-4}) and a final manual rotary step attenuator (At_{var}) used to sweep the receiver toward its sensitivity threshold. All components are matched to a 50Ω characteristic impedance.

The precision attenuators used were Mini-Circuits model VAT-30+, which provide a flat response within the 915 MHz band with a $VSWR \approx 1.2$. Interconnections were made using 50Ω SMA-to-SMA coaxial jumpers with measured insertion loss $L_{ins} < 0.5$ dB per segment. The combined uncertainty of the attenuator chain, including connector mismatch and component tolerances, is estimated at ± 0.8 dB across the measurement bandwidth.

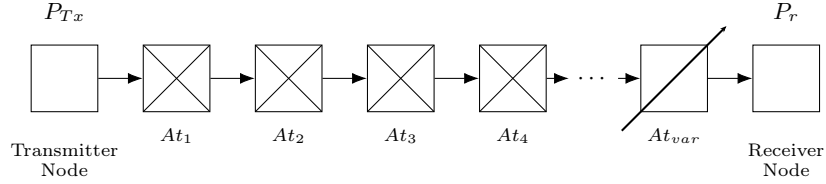


Fig. 1: Experimental guided-link system model using fixed attenuators (At_{1-4} : Mini-Circuits VAT-30+, 50Ω) and a manual rotary step attenuator (At_{var} : 0–110 dB).

The received power P_r is calculated as:

$$P_r = P_{Tx} - L_{ins} - \sum_{x=1}^n At_x \quad (1)$$

where P_{Tx} is the transmission power, L_{ins} represents the aggregate insertion losses of connectors and jumpers, and At_x denotes the attenuation of each stage.

The study utilised the Heltec WiFi LoRa 32 V3.2 and the RAK4631 Wis-Block as primary nodes. Both utilise the Semtech SX1262 LoRa chipset. It is important to note a discrepancy between software reporting and hardware limits: while the Meshtastic interface for the RAK4631 node used as a transmitter may allow settings up to +30 dBm, the SX1262 transceiver is physically limited to a maximum output of $+21 \pm 1$ dBm [10]. Consequently, this study adopts qualitative labels (Low, Medium, Max) to describe transmission power levels, where “Max” represents the hardware’s maximum effective saturation point. Table 1 compares the key technical specifications of both nodes.

3.2 LoRa Physical Layer and Meshtastic Presets

Modem Presets (SF, BW, CR). Meshtastic abstracts the complexity of LoRa modulation through “Modem Presets,” which are standardised combinations of three core variables:

Table 1: Technical Comparison of Meshtastic Nodes under Evaluation

Feature	Heltec V3.2	WiFi	LoRa	32 RAK4631	WisBlock
Master MCU	ESP32-S3FN8 (Dual-core LX7, 240 MHz)			Nordic nRF52840 M4F, 64 MHz)	(Cortex-
LoRa Chipset	Semtech SX1262			Semtech SX1262	
LoRa Sensitivity	-137 dBm (SF12 125 kHz)		(SF12 /	-137 dBm (SF12 125 kHz)	/
Max. TX Power	21 ± 1 dBm			22 dBm	
Flash / RAM	8 MB Flash / 512 kB SRAM			1 MB Flash / 256 kB RAM	
Onboard Display	0.96" OLED (128 × 64)			None (external via I ² C)	
USB Interface	Type-C (CP2102 bridge)			Type-C (Native USB)	

- **Spreading Factor (SF):** Defines the duration of the chirps. A higher SF (up to 12) increases the link budget and sensitivity, allowing signals to be decoded even under the noise floor, but significantly increases Time-on-Air (ToA).
- **Bandwidth (BW):** The frequency range of the signal. Reducing BW (e.g., from 500 kHz to 125 kHz) improves sensitivity at the cost of lower bitrates.
- **Coding Rate (CR):** The ratio of forward error correction bits. Higher CR (e.g., 4/8) increases redundancy and reliability in high-attenuation environments.

These presets allow the system to adapt to different mission requirements, ranging from high-speed local telemetry (*Short Turbo*) to maximum-range emergency messaging (*Long Slow*), as detailed in Table 2.

Table 2: Meshtastic Modem Presets and LoRa Parameters

Preset Name	BW (kHz)	SF	CR	Target
Short Turbo	500	7	4/5	High Speed
Short Fast	250	7	4/5	Balanced Local
Short Slow	250	8	4/5	Balanced Fast
Medium Fast	250	9	4/5	Balanced Mesh
Medium Slow	250	10	4/5	Robust Mesh
Long Fast*	250	11	4/5	Default Range
Long Moderate	125	11	4/8	High Range
Long Slow	125	12	4/8	Maximum Range

*Default configuration.

CSS Modulation and Sub-Noise-Floor Operation. The link behaviour is governed by Chirp Spread Spectrum (CSS) modulation. In urban canyon or

emergency rescue scenarios, signals often operate below the thermal noise floor ($SNR < 0$). LoRa transceivers like the SX1262 can demodulate signals with negative SNR, down to approximately -20 dB for high spreading factors [10]. This characteristic is fundamental for the experimental design, as the guided-link attenuator chain must be capable of driving the received signal well below the noise floor to map the full sensitivity curve.

SNR vs. RSSI: Critical Metric Selection. In high-attenuation environments, the Signal-to-Noise Ratio (SNR) becomes a more critical metric than the Received Signal Strength Indicator (RSSI). While RSSI measures total power at the receiver front-end, it loses granularity at the noise floor [9]. The actual received power (P_{rx}) is corrected as follows:

$$P_{rx} = \begin{cases} RSSI_{packet} + SNR_{packet}, & \text{if } SNR < 0 \\ RSSI_{packet}, & \text{if } SNR \geq 0 \end{cases} \quad (2)$$

This correction is particularly relevant for presets with high SF and narrow BW (e.g., *Long Slow*), where the receiver operates deep into the negative-SNR region before link failure occurs.

3.3 Spectral Verification of LoRa Emissions

To validate the spectral characteristics of the DUT before the guided-link experiments, the transmitted signal was captured using two independent measurement systems. This dual-instrument approach provides both time-frequency visualisation of the CSS modulation and a calibrated power spectral density reference.

Figure 2 shows the LoRa signal captured using an RTL-SDR receiver and the SDRangel software, centred at 902.125 MHz with a span of 3.2 MHz. The upper panel displays the real-time power spectral density, where two transmission peaks are visible around 901.9 MHz and 902.2 MHz, reaching approximately -20 dBm above a noise floor of -70 dBm to -80 dBm. The lower panel presents the spectrogram (waterfall display), where the characteristic diagonal chirp patterns of CSS modulation are clearly identifiable. These frequency sweeps, visible as diagonal bright traces, confirm that the LoRa transmitter is operating correctly with the expected modulation scheme.

Figure 3 presents the spectral measurement obtained with an Anritsu hand-held spectrum analyser, centred at 926.750 MHz with a 5 MHz span. The instrument was configured with a resolution bandwidth (RBW) of 300 kHz, video bandwidth (VBW) of 100 kHz, and a sweep time of 108 ms in Max Hold trace mode. The captured spectral envelope shows the accumulated LoRa signal occupying approximately 925 MHz to 928 MHz, with peak power levels near -35 dBm (accounting for 2.2 dB of external gain noted in the instrument settings). The occupied bandwidth and centre frequency are consistent with the 915 MHz ISM band channelisation used by Meshtastic in the Americas region (AU915/US915 frequency plan).

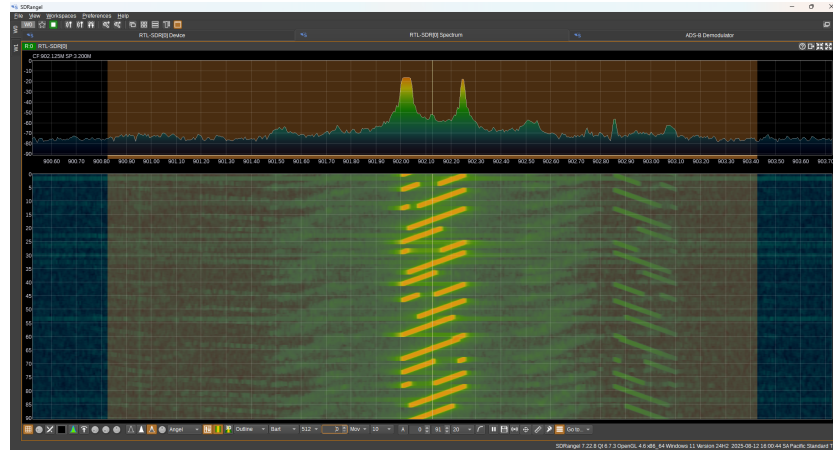


Fig. 2: LoRa signal captured with RTL-SDR and SDRangel at 902.125 MHz (span: 3.2 MHz). Upper: PSD showing transmission peaks. Lower: waterfall display with characteristic CSS chirp patterns.

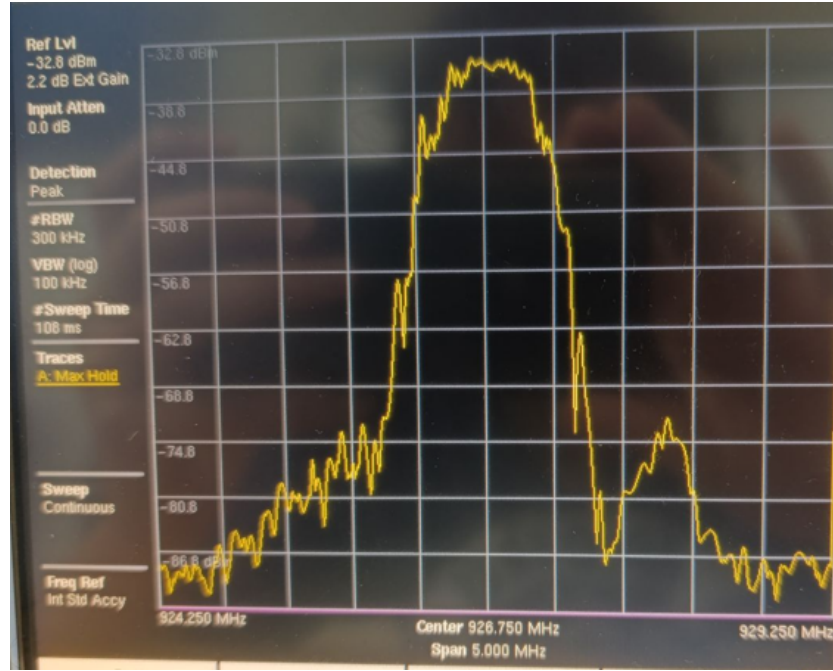


Fig. 3: LoRa spectral envelope captured with Anritsu spectrum analyser at 926.750 MHz (span: 5 MHz, RBW: 300 kHz, Max Hold). The occupied bandwidth confirms operation within the 915 MHz ISM band.

The combination of both measurements—the SDR waterfall confirming CSS chirp structure and the calibrated analyser verifying power levels and occupied bandwidth—provides robust evidence that the DUT emits signals conforming to the LoRa specification prior to entering the guided-link attenuator chain.

3.4 Experimental Design

Variable Classification. Table 3 explicitly categorises the experimental variables to ensure reproducibility.

Table 3: Experimental Variable Classification

Type	Variable	Values / Range
Independent	Modem Preset	8 presets (Table 2)
	TX Power Level	Low, Medium, Max
	At_{var}	0–110 dB (1 dB steps)
Dependent	RSSI	dBm
	SNR	dB
	PER	% (per 50-packet burst)
Controlled	Fixed attenuation	$\sum At_{1-4}$ (constant)
	Temperature	Ambient (20 °C to 25 °C)
	Packet payload	Fixed tag per step

To characterise performance across the Meshtastic operational spectrum, a comprehensive matrix of 42 datasets was executed. This consists of all 8 modem presets evaluated at three power levels, as summarised in Table 4. Presets with shorter ToA (*Short Turbo* through *Long Fast*) were measured in duplicate runs per power level to assess repeatability, yielding 36 datasets. Presets with longer ToA (*Long Moderate* and *Long Slow*) were measured once per power level due to extended acquisition times, contributing 6 additional datasets.

The variable attenuation (At_{var}) was incremented manually using a rotary step attenuator (0–110 dB in 1 dB steps). For each attenuation step, a burst of 50 packets was transmitted. After each burst, a stabilisation period of 5 s was observed before incrementing the attenuator to the next step. Data was captured through the CP2102 serial interface at 115 200 baud and logged to CSV files, recording timestamp, RSSI, SNR, and a “payload” tag correlated with each attenuation step.

The Packet Error Rate (PER) was calculated for each attenuation step as:

$$PER = \frac{N_{lost}}{N_{sent}} \times 100\% \quad (3)$$

where $N_{sent} = 50$ packets per step and N_{lost} is the number of packets not received or received with CRC errors. The *sensitivity threshold* for each modem

Table 4: Experimental Test Matrix (42 Datasets Total)

Meshtastic Preset	Power Levels	Runs/Level	Datasets
Short Turbo	Low, Med, Max	2	6
Short Fast	Low, Med, Max	2	6
Short Slow	Low, Med, Max	2	6
Medium Fast	Low, Med, Max	2	6
Medium Slow	Low, Med, Max	2	6
Long Fast	Low, Med, Max	2	6
Long Moderate	Low, Med, Max	1	3
Long Slow	Low, Med, Max	1	3
Total			42

preset-power combination was defined as the minimum received power P_r at which $PER \leq 10\%$. Beyond this threshold, the link is considered unreliable for operational Meshtastic deployment.

4 Results and Analysis

This section presents the experimental results obtained from the 42-dataset test matrix. The analysis focuses on the joint behaviour of RSSI and SNR as a function of guided-link attenuation across all eight Meshtastic modem presets at maximum transmission power (effective +21 dBm).

Figures 4 present the complete measurement results:

4.1 Spreading Factor and Bandwidth Impact on Sensitivity

The RSSI data in Fig. 4(a) reveals distinct behavioural families, separated primarily by their Spreading Factor (SF). The *Short* family ($SF7$ – $SF8$, $BW \geq 250$ kHz) exhibits a quasi-linear RSSI decay of approximately -1 dB per 1 dB of added attenuation, consistent with the expected response of a calibrated guided link. *Short Turbo* ($SF7$, $BW = 500$ kHz), having the lowest processing gain, reaches the receiver noise floor first, saturating near -90 dBm at approximately 75 dB. *Short Fast* ($SF7$, $BW = 250$ kHz) and *Short Slow* ($SF8$, $BW = 250$ kHz) collapse progressively between 110 dB to 120 dB.

The *Medium* family ($SF9$ – $SF10$, $BW = 250$ kHz) extends reliable reception into the 135 dB to 150 dB range, with the additional processing gain of $SF9$ and $SF10$ providing approximately 20 dB to 30 dB of additional margin over the *Short* presets. The high processing gain *Long* family ($SF11$ – $SF12$) demonstrates markedly extended range. Even at a wider bandwidth, *Long Fast* ($SF11$, $BW = 250$ kHz) survives up to 155 dB. *Long Slow* ($SF12$, $BW = 125$ kHz) maximises the link budget, maintaining packet reception up to approximately 180 dB of total attenuation. This additional margin is attributable to the 3 dB sensitivity improvement from halving the bandwidth combined with the maximum CSS processing gain of $SF12$.

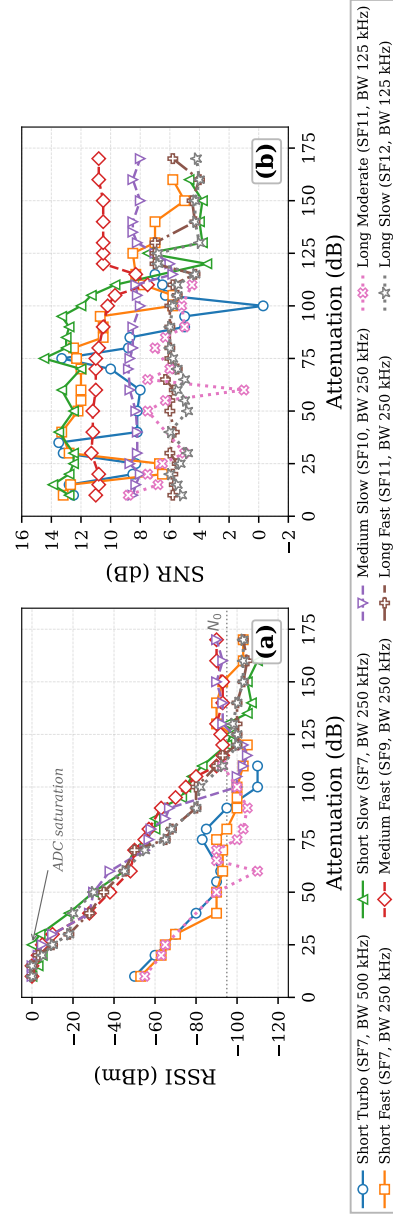


Fig. 4: Comparison of Meshtastic modem presets: (a) Received Signal Strength Indicator (RSSI) vs. Path Attenuation and (b) Signal-to-Noise Ratio (SNR) vs. Path Attenuation. The dashed line N_0 indicates the receiver thermal noise floor.

4.2 SNR Stratification and Sub-Noise-Floor Demodulation

The SNR data in Fig. 4(b) reveals a clear stratification by spreading factor. At low attenuation, the presets cluster into tiers: a high tier (12 dB to 14 dB) occupied by the $SF7$ – $SF8$ presets (*Short Turbo*, *Short Fast*, and *Short Slow*), a mid tier (8 dB to 11 dB) corresponding to $SF9$ – $SF11$ at $BW = 250$ kHz (*Medium Fast*, *Medium Slow*, *Long Fast*), and a lower initial tier (5 dB to 7 dB) for the narrowband $BW = 125$ kHz presets (*Long Moderate*, *Long Slow*).

A defining characteristic of the LoRa physical layer is explicitly captured in this dataset: the ability to demodulate signals deeply embedded in thermal noise. As attenuation increases beyond 120 dB, the SNR for high-SF presets crosses the zero-threshold and descends linearly into the negative regime. *Long Slow* reaches absolute SNR values of approximately -18 dB before link failure, demonstrating that the CSS processing gain effectively preserves demodulation capability near the SX1262’s theoretical limit (-20 dB for $SF12$). The data also suggests that link failure for high-SF presets is *abrupt* rather than gradual. The transition from reliable reception to complete link loss occurs while the system still maintains a tight 2 dB to 4 dB margin above the absolute theoretical demodulation limit, confirming that fade margins in the order of 5 dB are required for stable network planning.

4.3 Measurement Artefacts

Two systematic artefacts were identified in the dataset, providing important bounds for signal interpretation.

Front-End ADC Saturation. Several presets report $RSSI \approx 0$ dBm at low attenuation levels (0 dB to 40 dB), visible in the upper-left region of Fig. 4(a). This indicates ADC saturation: the received signal exceeds the dynamic range of the SX1262 receiver, causing the RSSI register to clip. These data points do not represent actual propagation loss and must be excluded from path-loss regression models.

RSSI Plateau at the Noise Floor. Beyond the sensitivity limit, all surviving presets exhibit an RSSI plateau near -100 dBm that persists across extreme attenuation levels. In this regime, the RSSI register reflects the receiver’s thermal noise power rather than the signal energy. Consequently, the corrected received power must be calculated using the SNR-adjusted formula (Eq. 2), validating the theoretical prediction that SNR is the sole reliable link quality metric in the sub-noise-floor operating regime.

4.4 Anomalous Behaviour of Long Moderate

The *Long Moderate* preset ($SF11$, $BW = 125$ kHz, $CR = 4/8$) exhibits a highly anomalous discontinuity. As seen in Figs. 4(a) and 4(b), the link fails abruptly

near 80 dB of attenuation. Crucially, at the point of failure, the preset still maintains an extraordinarily robust SNR of approximately +10 dB. Because this abrupt drop does not follow the physical layer progression seen in *Long Slow* (*SF12*, same BW) nor *Long Fast* (*SF11*, wider BW), it strongly suggests a specific interaction between *SF11*, $CR = 4/8$, and the firmware’s Channel Activity Detection (CAD) or packet timeout mechanisms, rather than a true RF sensitivity limitation.

5 Discussion

The experimental results presented in Section 4 provide the empirical foundation for selecting the appropriate Meshtastic configuration according to the deployment scenario. This section analyses how the choice of modem preset directly determines the achievable inter-node distance and, consequently, the node density required to guarantee mesh connectivity over a given coverage area.

5.1 Operational Regimes and Configuration Selection

Three operational regimes can be identified from the sensitivity thresholds:

- **High-density, low-latency applications** (*Short Turbo*, *Short Fast*, *Short Slow*): suitable for dense urban IoT sensor networks, building automation, or campus-scale telemetry where nodes are closely spaced (<500 m) and frequent data updates are required. The high throughput and low ToA minimise channel occupancy per transmission, enabling a larger number of nodes to coexist within the same mesh without excessive collision probability. The *SF7–SF8* presets tolerate 110 dB to 120 dB of path loss, which in a typical dense urban environment (path loss exponent $n \approx 3.5\text{--}4$) corresponds to inter-node distances in the order of 100 m to 500 m.
- **Balanced urban mesh** (*Medium Fast*, *Medium Slow*, *Long Fast*): appropriate for neighbourhood-scale smart city deployments—environmental monitoring, traffic sensing, public safety alerts—where moderate inter-node distances (0.5 km to 2 km) balance coverage and throughput. *Long Fast*, as the default Meshtastic configuration, represents the most versatile trade-off for general-purpose urban mesh networks, sustaining links up to 155 dB of attenuation. *Medium Fast* is notable for maintaining a uniform SNR margin of approximately 10 dB across its operational range, making it particularly robust against temporal fading variations.
- **Maximum-range emergency networks** (*Long Slow*): designed for disaster response, rural coverage extension, or search-and-rescue operations where inter-node distances may exceed 2 km to 10 km. The reduced throughput and high ToA are acceptable trade-offs when the primary objective is to establish any communication link over the largest possible area with the minimum number of available nodes. The 180 dB link budget of *Long Slow* enables theoretical LOS ranges exceeding 10 km even with conservative fade margins.

It should be noted that the anomalous behaviour of *Long Moderate* (Section 4.4) currently limits its practical applicability despite its favourable theoretical parameters. Until the root cause—likely firmware-related—is resolved, *Long Slow* remains the recommended preset for maximum-range deployments.

5.2 Node Density and Economic Implications

The 60 dB to 70 dB span between the *Short Turbo* and *Long Slow* sensitivity thresholds translates directly into node density requirements for a given coverage area. Table 5 illustrates the estimated inter-node distances for a representative dense urban scenario.

Table 5: Estimated Inter-Node Distance and Density for Dense Urban Deployment ($n = 3.5$, $P_{Tx} = +21$ dBm)

Regime	Max Att. (dB)	Est. Range (m)	Nodes/km ² (approx.)
Short (SF7–8)	110–120	200–400	25–80
Medium (SF9–10)	135–150	500–1500	3–12
Long (SF11–12)	155–180	1500–5000	1–3

Ranges assume $f = 915$ MHz, free-space + urban excess loss.

Actual values depend on building density and terrain.

A *Short Turbo* deployment covering 1 km² may require 25 to 80 nodes, while the same area under *Long Slow* could be served by as few as one to three strategically placed nodes—at the cost of message latency increasing from seconds to minutes. For a municipal risk management agency planning a city-wide early warning system, this represents a trade-off between hardware investment (more nodes, lower unit cost per message) and operational simplicity (fewer nodes, higher per-node criticality).

The guided-link sensitivity thresholds established in Section 4 provide the baseline link budget from which planners can subtract the expected path loss—using models such as ITU-R P.1411 for urban microcells or the Okumura-Hata model for wider areas—to estimate the maximum inter-node spacing for each preset in a specific urban morphology. For Latin American cities, where building construction (reinforced concrete, brick masonry) introduces additional NLoS penetration losses at 915 MHz, a conservative margin of 10 dB to 15 dB beyond the measured thresholds is recommended.

6 Conclusions

This work presented a guided-link experimental methodology for the deterministic characterisation of Meshtastic nodes operating in the 915 MHz ISM band. By replacing the wireless channel with a calibrated attenuator cascade, the study

isolated the LoRa physical layer from stochastic propagation effects, enabling repeatable sensitivity and PER measurements across the full operational envelope of the SX1262 transceiver.

The evaluation of all eight modem presets at three power levels (42 datasets) revealed three distinct performance tiers governed primarily by Spreading Factor: the *Short* family ($SF7$ – $SF8$) failed between 110 dB to 120 dB, the *Medium* family ($SF9$ – $SF10$) sustained links up to 135 dB to 150 dB, and the *Long* family ($SF11$ – $SF12$) reached 180 dB—a 60 dB to 70 dB advantage that translates to multiple orders of magnitude in propagation distance. The SNR analysis confirmed sub-noise-floor demodulation down to -18 dB for $SF12$, with abrupt link failure within 2 dB to 4 dB of the theoretical limit, establishing that fade margins of 5 dB suffice for stable network planning.

These thresholds enabled the identification of three operational regimes—high-density IoT, balanced urban mesh, and maximum-range emergency—each with distinct node density and latency characteristics (Section 5). The analysis provides network designers and municipal risk management agencies with quantitative criteria to dimension Meshtastic deployments according to coverage requirements, budget constraints, and urban morphology.

Two measurement artefacts were documented (ADC saturation and noise floor plateau), and an anomalous firmware-level failure of the *Long Moderate* preset was identified, warranting further investigation.

Meshtastic represents a vital convergence between the sub-noise-floor capabilities of SX1262 transceivers and the need for communication resilience in smart cities. For Latin American urban contexts—where licence-free 915 MHz operation, low hardware cost, and autonomous power supply enable rapid deployment—the configuration-dependent design guidelines presented here constitute a practical planning framework for off-grid emergency and IoT networks.

6.1 Future Work

The results open several research directions:

1. **Outdoor propagation validation:** Field measurements in representative urban environments (dense urban, suburban, urban canyon), with particular emphasis on NLoS penetration losses through Latin American building materials at 915 MHz.
2. **Flooding protocol scalability:** Evaluation of Meshtastic’s managed flooding in dense NLoS scenarios, and exploration of hybrid routing strategies (selective forwarding, hierarchical clustering) for formal adoption by risk management entities.
3. **Node density optimisation tool:** Development of a planning tool mapping empirical sensitivity thresholds to optimal node placement for a given urban morphology, terrain, and QoS requirements.
4. **IoT platform integration:** Benchmarking of dual-mode operation (autonomous mesh / cloud-connected) with urban management platforms under realistic multi-hop traffic loads.

5. **Dynamic channel emulation:** Extension of the guided-link methodology to include multipath fading profiles and Doppler spread via hardware channel emulator.

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